BANKING RISK

ANALYTICS & CUSTOMER PORTFOLIO DASHBOARD

An end-to-end exploratory analytics project using Python, MySQL and Power BI

DATASET	FIELDS	TOOLS	OUTPUT
3,000 records	26 prepared columns	Python • MySQL • Power BI	Interactive dashboard

1. Executive Summary

This project explores how customer, demographic and financial information can be organized to understand a bank's customer portfolio, lending exposure and deposit base. The workflow combines Python-based exploratory data analysis (EDA), relational data preparation and querying in MySQL, and interactive reporting in Power BI. The resulting dashboard presents key metrics and allows users to examine loans, deposits and customer segments through filters and visual summaries.

2. Business Context and Problem Statement

Banks manage several types of financial exposure across customers, including bank loans, business lending and credit card balances. Reviewing these amounts alongside estimated income, demographics, account balances and banking relationships can help analysts understand portfolio composition and identify segments that may require closer review. The project therefore focuses on descriptive risk and portfolio analytics: making customer and financial information easier to explore and compare.

3. Project Objectives

- Inspect and prepare customer-level banking data for analysis.
- Explore distributions and relationships across demographic and financial variables using Python.
- Build a relational MySQL structure, validate keys and create a joined analysis view.
- Develop Power BI measures and interactive pages for loan, deposit and customer portfolio analysis.

Analysis workflow

01	02	03	04	05
Python EDA	MySQL setup	SQL joins	Power BI model	Dashboard

DATA UNDERSTANDING

4. Dataset, Preparation and Exploratory Analysis

The analysis uses a prepared customer-level banking dataset containing customer attributes and financial product balances.

4.1 Dataset Overview

The prepared analysis file contains 3,000 customer records and 26 columns. Variables cover customer identifiers, age, location, date of joining, nationality, occupation, fee structure, loyalty classification, estimated income, superannuation savings, credit cards, loans, deposits, account balances, properties owned, risk weighting, banking relationship, gender and investment advisor. The data is organized to support customer segmentation and financial portfolio analysis.

CHECK	RESULT	HOW IT WAS USED
Prepared dataset size	3,000 rows × 26 columns	Established the analysis scope.
Missing values	No null values reported in the notebook checks	Supported the initial EDA without a null-imputation step.
Exact duplicate rows	0 duplicate rows	No fully duplicated records were identified.
Repeated ClientID values	60 repeated occurrences; 2,940 distinct ClientID values	A separate surrogate customer_key was used as the row-level primary key.
Date field	Joined Bank converted to datetime	Enabled tenure and engagement-duration analysis.

4.2 Data Preparation

- Loaded the prepared customer analysis file into pandas and reviewed its shape, column names, data types and descriptive statistics.
- Converted Joined Bank into a datetime field so that customer tenure could be examined consistently.
- Created income bands using estimated income: Low (up to 100,000), Mid (above 100,000 up to 300,000), and High (above 300,000). These categories support grouped comparisons in the dashboard.
- Reviewed categorical and numeric variables, including gender, nationality, banking relationship, income, loans, deposits and account balances.

4.3 Exploratory Data Analysis

EDA used descriptive statistics and visual analysis to understand the dataset before creating the dashboard. Categorical variables were examined with value counts and bar charts. Numeric variables were explored with histograms, while a correlation heatmap and selected scatter/regression plots were used to study relationships among financial measures.

OBSERVATION	WHAT THE DATA SHOWED
Gender distribution	The dataset is almost evenly split: 1,512 female customers and 1,488 male customers.
Banking relationship	Private Bank is the largest segment (1,352 customers), followed by Retail (660), Institutional (495) and Commercial (493).
Nationality	European customers are the largest nationality group in the prepared dataset (1,309 records across five nationality categories).
Account balances	The correlation analysis indicated positive relationships among bank deposits, checking accounts, savings accounts and foreign currency balances.
Other financial variables	The analysis explored income, superannuation, bank loans, credit card balances, business lending and properties owned; correlations varied by variable pair.

RELATIONAL DATA PREPARATION

5. MySQL Database Design and SQL Analysis

MySQL was used to structure the customer data, enforce key relationships and prepare a reusable joined view for downstream analysis.

5.1 Database Structure

A database named bank was created. The SQL workflow used four tables: customer, bank_relations, gender and advisors. The customer table stores the main customer and financial attributes, while the other tables contain descriptive lookup values for banking relationship, gender and investment advisor. This separation makes the relationships explicit and allows descriptive attributes to be brought into the analysis through joins.

TABLE	KEY USED	ROLE IN THE ANALYSIS
customer	customer_key (surrogate PK); BRId, GenderId, IAId (FKs)	Main customer and financial information.
bank_relations	BRId (PK)	Maps relationship IDs to relationship categories.
gender	GenderId (PK)	Maps gender IDs to readable gender labels.
advisors	IAId (PK)	Maps advisor IDs to investment advisor names.

5.2 Key Validation and Duplicate Handling

Initial checks examined row counts, column definitions, null ClientID values and duplicate identifiers. The customer table contained repeated ClientID values: 60 additional occurrences beyond the 2,940 distinct IDs. Since ClientID was therefore not unique at the row level, an auto-increment customer_key column was added and used as the primary key. The lookup tables were assigned primary keys on BRId, GenderId and IAId, and the corresponding columns in customer were configured as foreign keys.

5.3 Joins, Aggregations and Analysis View

A series of LEFT JOIN operations connected customer to bank_relations, gender and advisors. LEFT JOIN was selected so customer records remained available even if a matching descriptive lookup value was absent. The joined query selected customer identifiers, demographics, banking dates, financial balances and readable relationship, gender and advisor labels.

```
SELECT c.customer_key, c.ClientID, c.Name, c.Age,
       br.`Banking Relationship`, g.Gender, a.`Investment Advisor`,
       c.`Estimated Income`, c.`Bank Loans`, c.`Bank Deposits`
FROM customer c
LEFT JOIN bank_relations br ON c.BRId = br.BRId
LEFT JOIN gender g ON c.GenderId = g.GenderId
LEFT JOIN advisors a ON c.IAId = a.IAId;
```

The joined query was also used to verify the resulting row count. Aggregation queries summarized customers by gender and banking relationship, and calculated total bank loans by relationship category. Finally, the joined query was saved as a reusable SQL view named customer_analysis, allowing the prepared data to be queried without repeating the joins.

SQL deliverables

Database creation and inspection queries • row-count and duplicate checks • primary-key and foreign-key definitions • multi-table LEFT JOIN • grouped customer and loan aggregations • customer_analysis view.

BUSINESS INTELLIGENCE LAYER

6. Power BI Data Model, DAX Measures and KPIs

Power BI converts the prepared customer data into interactive measures and pages for customer, loan and deposit analysis.

6.1 Data Transformation and Calculated Columns

The joined customer data was used in Power BI as the Clients - Banking analysis table. Additional fields were created to make the dashboard easier to segment and interpret:

- Engagement Timeframe: groups customers by the duration or period of their relationship with the bank.
- Engagement Days: calculates the number of days from Joined Bank to the current date using DATEDIFF.
- Income Band: groups Estimated Income into Low, Mid and High categories using the thresholds described in the EDA section.
- Processing Fees: derives a fee rate from Fee Structure. The documented project rule includes a 5% rate for the High fee category; other configured category rates should be interpreted according to the Power BI file.

6.2 Core DAX Measures

MEASURE	PURPOSE	DAX LOGIC
Total Clients	Distinct customer count.	DISTINCTCOUNT(Client ID)
Bank Loan	Sum of bank loan balances.	SUM(Bank Loans)
Business Lending	Sum of business lending balances.	SUM(Business Lending)
Credit Card Balance	Sum of card balances.	SUM(Credit Card Balance)
Total Loan*	Combined lending/credit exposure used in this dashboard.	[Bank Loan] + [Business Lending] + [Credit Cards Balance]
Bank Deposit	Sum of bank deposits.	SUM(Bank Deposits)
Total Deposit	Combined deposit balances across account types.	[Bank Deposit] + [Savings Account] + [Foreign Currency Account] + [Checking Accounts]
Total Fees	Estimated fees using the configured rate.	SUMX(Clients - Banking, [Total Loan] × [Processing Fees])
Total Credit Cards	Total number of cards held.	SUM(Amount of Credit Cards)
Engagement Days	Customer tenure in days.	DATEDIFF(Joined Bank, TODAY(), DAY)

6.3 Dashboard Structure

The Power BI report is organized into four pages. KPI cards provide a high-level overview, while charts and slicers let users compare the portfolio across customer segments. Page navigation supports movement between the main analysis areas.

INSIGHTS AND CONCLUSION

7. Dashboard Outputs, Findings and Future Enhancements

The final report brings the analysis together in a format intended for exploratory business review.

DASHBOARD PAGE	CONTENT AND USE
Home	Overview cards for total clients, combined loan exposure, deposits and fees, with filters and navigation.
Loan Analysis	Breakdown of loan and credit exposure by banking relationship, income band, nationality and occupation.
Deposit Analysis	Comparison of bank deposits, savings, checking and foreign currency balances across customer segments.
Summary Dashboard	Consolidated view of key banking metrics for quick review.

7.1 Key Findings and Business Relevance

- Private Bank customers form the largest relationship segment in this sample, making relationship type a useful dimension for portfolio comparisons.
- The gender split is balanced, while nationality distribution is led by European customers in the supplied data.
- The positive associations observed among several deposit and account-balance variables suggest that customers with higher balances in one account category may also hold higher balances in other deposit categories.
- The combined loan and deposit measures allow users to compare financial exposure and deposit holdings across income bands, nationalities, occupations and banking relationships.

7.2 Limitations

- The dataset does not include a confirmed repayment/default outcome, credit score history or loan performance label. Default probability and model accuracy therefore cannot be evaluated.
- The analysis is based on a prepared sample dataset; results may not represent a live bank portfolio or the wider banking market.
- The combined loan, total deposit and processing-fee calculations are project-defined analytical measures and should be validated with banking domain or finance stakeholders before operational use.

7.3 Future Enhancements

- Integrate repayment history, overdue amounts, delinquency status and credit bureau variables to enable a properly labelled credit-risk analysis.
- Build and validate a predictive model only after defining a reliable target variable, preventing data leakage and evaluating performance with suitable metrics.
- Add time-based portfolio trends, cohort analysis, outlier monitoring and more detailed customer-level drill-through views.
- Validate fee-rate mappings and all aggregate KPI definitions against documented business rules.

8. Conclusion

This project demonstrates a practical analytics workflow across Python, MySQL and Power BI. Python was used to inspect the data, explore distributions and examine relationships; MySQL provided relational structure, key validation, joins, aggregations and a reusable analysis view; and Power BI presented customer, loan and deposit metrics through an interactive multi-page dashboard. The result is an exploratory portfolio analytics solution that helps organize banking information and surface patterns for further investigation. It also provides a foundation for extending the work into more advanced credit-risk analysis.

Technologies: Python • pandas • Matplotlib • Seaborn • MySQL • SQL • Power BI • DAX